Detailed Project Report (DPR)

Tender-Coconut & Agri-Biomass Biochar and Enriched Bio-Fertilizer Unit

Location: Eruthenpathy, Chittur Taluk, Palakkad District, Kerala (Kerala–Tamil Nadu border) **Feedstock origin:** Tender-coconut husk briquettes from a partner packaging unit in the regional industrial belt + own coconut farm residues **Prepared for:** Submission to District Industries Centre (DIC), Palakkad and lending bank (AIF / term loan) **Document status:** Planning-grade DPR — figures are indicative estimates to be firmed up with (a) a feedstock proximate-analysis lab report and (b) formal machinery quotations before sanction.

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1. Executive summary

The project sets up a **continuous, low-emission biochar plant with an enriched (microbe-inoculated) organic bio-fertilizer line** on an owned **38-cent plot**, using **4 tonnes/day of tender-coconut husk briquettes** (from a partner tender-coconut packaging unit ~30–40 km away) as the anchor feedstock, supplemented by coconut fronds, shells and coir pith from the promoters' adjacent farm.

- **Phase-1 capacity: ~1 tonne/day biochar** (~300 t/yr) — matched to the 4 t/day briquette feed.
- **Phase-2 (expansion): ~2 tonnes/day biochar** (~600 t/yr) by adding a second modular reactor and aggregating farm/regional biomass.
- **Design philosophy:** all **fixed, permitted and civil infrastructure is built for the ultimate 2 t/day**, while **modular equipment (reactor) is bought for 1 t/day and expanded later** as a bolt-on — no re-permitting or civil rework.
- **Total Phase-1 project cost: ~₹1.70 crore**.
- **Financing:** promoter equity + **AIF term loan (3% interest subvention, 7 yrs)** + **CGTMSE** collateral-free cover + **Kerala ESS capital subsidy (~₹40 lakh)**.
- **Revenue stack:** enriched biochar sales (domestic + export) + **biochar carbon-removal credits** via an aggregator (Puro.earth CORCs).
- **Indicative payback:** ~3–4 years (physical-sales base case); under ~2 years if carbon-credit revenue is realized.

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2. Promoter

Promoter entity: *Arun and Friends* (proposed partnership enterprise).

- Promoted by Arun and associates ("Arun and Friends").
- Owns the project site (38-cent plot) and an adjacent coconut farm (~200 m away) in Eruthenpathy, Palakkad.
- Anchor feedstock (~4 t/day tender-coconut husk briquettes) is secured from a partner tender-coconut packaging unit in the regional industrial belt under a supply arrangement — the residue is currently a disposal cost and becomes a secured, low-cost feedstock.
- Captive demand: the promoters' own coconut farm can consume and demonstrate the enriched product, de-risking market entry.

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3. Project concept & rationale

Tender-coconut husk is a wet, fibrous, **potassium-rich** residue; converting it (and other coconut/agri biomass) to biochar via controlled pyrolysis: 1. Solves the partner's daily waste-disposal problem (secured feedstock). 2. Produces a stable, carbon-rich soil amendment; enriched with organics + microbes it becomes a branded slow-release bio-fertilizer. 3. Generates durable **carbon-removal credits** (1 t biochar ≈ ~2.2 t CO_2 sequestered for centuries). 4. Qualifies for strong central + state subsidy support as an agro-based / bio-fertilizer MSME.

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4. Market analysis

Product (physical biochar), indicative prices:

Grade	₹/kg	25 kg bag
Raw bulk (agri)	15–40	₹375–1,000

Grade	₹/kg	25 kg bag
Enriched / inoculated	50–120	₹1,250–3,000
Premium certified (EBC/organic)	up to 150	up to ₹3,750

Carbon credits: biochar CORCs trade ~US$125–145 (Nasdaq CORCCHAR index); Indian industrial biochar clears ~€105–150/t CO_2. Biochar delivered ~43% of verified durable CDR volume; >90% of near-term supply is already contracted by buyers (Microsoft, Google, JPMorgan, Swiss Re) — a **supply-constrained, seller-favourable market**.

Demand strategy: anchor demand on (a) own-farm consumption, (b) enriched/branded regional sales, (c) export of the carbon credit via an aggregator. Sell *enriched* product (margin) rather than raw char (commodity).

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5. Feedstock plan (two-site model)

Site	Role
Farm (200 m from plot)	Feedstock aggregation, **sun-drying yard**, chipping/shredding, bulk storage. No processing here.
38-cent plot	Registered unit: pyrolysis + emission control + enrichment + product storage + office/lab.

Feedstock sources: partner's tender-coconut husk briquettes (4 t/day anchor); coconut fronds/petioles, shells, spathe from own farm; coir pith and shells from local dehuskers/coir units; seasonal agri residue (Phase-2).

Benefits of the split: sun-drying at the farm supplies free drying energy and reduces transported water; keeps large fuel storage away from the reactor (fire-safety and consent advantage); frees plot space so the 38 cent comfortably hosts the 2 t/day endgame.

Supply security: execute feedstock supply MoUs (partner + 2–3 local sources) — required for both bank appraisal and carbon-credit additionality/traceability.

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6. Technical design & capacity

6.1 Mass balance (design basis: 4 t/day briquettes, medium moisture ~18%)

Parameter	Value
Briquette feed	4,000 kg/day
Dry matter (~82%)	~3,280 kg/day
Biochar yield (~30% of dry mass)	**~1.0 t/day** (0.9–1.05 at 12–25% moisture)
Operating basis	~200 kg/hr feed → ~50 kg/hr biochar over ~20 h
Annual biochar (300 days)	**~300 t/yr (Phase-1)**
CO_2 removal (~2.2×)	~660 t CO_2/yr (Phase-1)

Phase-2: add a second identical reactor + aggregate biomass to ~8 t/day feed → **~2 t/day biochar (~600 t/yr, ~1,320 t CO_2/yr).**

6.2 Expandable-design split ("small where scalable, big where fixed")

Component	Sized now	Rationale
Pyrolysis reactor / carbonizer	**1 t/day (modular)**	Add 2nd parallel unit later; avoids idle capacity
Char cooling, sieving, bagging, enrichment mixer	1 t/day	Cheap to duplicate
Afterburner/flare, cyclone, scrubber, ID fan, stack	**2 t/day (big)**	Re-engineering gas train forces re-permitting
KSPCB CTE/CTO consent capacity	**Apply for 2 t/day now**	Consent amendment later is slow/painful
Electrical transformer, sanctioned load, panels	**2 t/day (big)**	Load enhancement/re-cabling costly
Civil: plinths, sheds, roads, foundations	**2 t/day (big)**	Pour concrete once; leave bolt pockets for reactor #2
Common pre-bin, waste-heat dryer, feed headers	**2 t/day (big)**	Pre-fit flanged tie-in stubs
Weighbridge, water, drainage, fire system, greenbelt	**2 t/day (big)**	One-time infrastructure
Farm drying floor + storage	For ~8 t/day feed	Low-cost area

Key move: install one reactor now but pre-provision foundation, gas/electrical/feed tie-ins and spare panel ways for reactor #2 → expansion is a bolt-on with minimal downtime and no re-permitting.

6.3 Emission control (for KSPCB CTO — Orange category)

Syngas afterburner/flare (self-fuels the process + drying) → cyclone/dust collector → alkaline scrubber with mist eliminator → stack meeting KSPCB ambient limits. Real-time temperature monitoring; emergency shutdown.

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7. Machinery & suppliers

Local (Coimbatore–Pollachi hub, 30–60 km — preferred for service/spares; build for coconut feedstock): coconut-shell/biomass carbonization plant manufacturers (e.g., Micro Fab Engineers, Essar Engineers, Veera Biopower and similar). Indicative: 5 t/day ≈ ₹66–72 L; 1–5 t/day semi-auto ≈ ₹6–25 L.

National (continuous, low-emission, certification-grade — better for CTO/EBC/carbon): - **Kerone Engineering (Mumbai)** — continuous multi-feedstock pyrolysis, waste-heat drying, cyclone + scrubber, SCADA. - **W2E Innovations (Ahmedabad/Kharagpur; Bengaluru branch)** — integrated biomass→biochar + wood-vinegar + fertilizer, "verification-ready." - **Susstains Engineering (Karnataka)** — IIT-Madras-derived zero-smoke continuous process (~34% yield). - **Ylem Energy (New Delhi)** — smaller batch furnaces (lower cost; batch → less consistent, weaker for CTO/EBC).

Recommendation: buy a **continuous/semi-continuous plant with integrated dryer + afterburner + scrubber** (not a bare batch kiln). Obtain quotes from one local Coimbatore/Pollachi maker (price + service) and one national continuous-tech supplier; compare on emission compliance and yield. Budget ₹40 L machinery for Phase-1 with the gas train and dryer pre-sized for 2 t/day.

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8. Land, layout & civil works

Plot: owned **38 cent = 16,553 sq ft = ~1,538 sq m**. Machine footprint is small; feedstock storage is off-site at the farm, so the plot comfortably hosts the 2 t/day endgame.

Indicative layout (2 t/day-ready):

Zone	Area (sq m)
Greenbelt / setback (KSPCB)	~550
Pyrolysis + emission control + waste-heat finish-dryer	150
Feedstock day-bin (1–2 day buffer)	80
Enrichment / composting + curing	250
Product storage + bagging	150
Office / lab / weighbridge / utilities	100
Internal roads / loader turning	~258
Total	**~1,538 (= 38 cent)**

Pre-checks: confirm plot land-use is convertible to industrial (Orange category needs land-use + Panchayat permission); KSPCB setback from residences/water bodies/eco-sensitive zones; road access for tippers; 3-phase power; water availability.

Construction (sized for 2 t/day): enclosed sheds ~35–40 L; roofed open sheds ~27–32 L; site works (leveling, compound wall, paving, drainage) ~10–14 L; ≈ **₹65–90 L** (see cost table).

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9. Regulatory & statutory compliance

Approval	Authority	Notes
Udyam (MSME) registration	Online	Gateway for all schemes
Consent to Establish (CTE)	KSPCB	Before construction; **apply rated for 2 t/day**
Consent to Operate (CTO)	KSPCB	After build + inspection; Orange category
Panchayat D&O (Dangerous & Offensive) licence	Grama Panchayat	Manufacturing unit
Building permit	Grama Panchayat	Sheds/plant
Fire NOC	Fire & Rescue Dept.	Thermal process
Factories Act registration	If worker/power thresholds crossed	
GST registration		
Land-use conversion	Revenue Dept.	If plot is agricultural

Operating without CTE/CTO is a criminal offence under EPA 1986. No open burning of biomass at any stage.

10. Cost of project (Phase-1)

#	Item	₹ Lakh
1	Land (owned 38 cent — promoter contribution in kind)	15
2	Site development, compound wall, roads, greenbelt	10
3	Civil — sheds & buildings (sized for 2 t/day)	48
4	Pyrolysis reactor — 1 t/day (Phase-1, modular)	20
5	Dryer + emission control + gas train (sized 2 t/day)	18
6	Sieving, mixing, pelletizer, bagging	6
7	Electrical infra & transformer (sized 2 t/day)	12
8	Weighbridge, water (borewell/tanks), fire system	13
9	Farm-side drying floor, chipper/shredder, storage	9
10	Lab / QC equipment, office	5
11	Preliminary & pre-operative (CTE/CTO, DPR, MRV setup, EBC prep)	7
12	Contingency (~5%)	7
	Fixed capital sub-total	**170**
13	Margin money for working capital	15
	Total project cost (Phase-1)	**~₹1.70 crore**

Phase-2 add-on (later): second reactor + handling ≈ ₹25–35 L, funded from cash flows / top-up loan.

11. Means of finance & subsidy stack

Source	₹ Lakh	%
Promoter equity (cash ~₹30 L + land in-kind ~₹15 L)	45	26%
Term loan — bank, under **AIF (3% interest subvention, 7 yrs)**, **CGTMSE** collateral-free	110	65%
Working capital limit (bank cash credit)	15	9%
Total	**170**	100%

Kerala ESS capital subsidy (~₹40 L) is received as **reimbursement after commissioning** and will be used to **prepay the term loan**, reducing effective debt to ~₹70 L.

Subsidy stack logic

- **Anchor = Kerala ESS** (state capital subsidy). Target **45%** band: 25% base (young/women/SC-ST/NRK) + 10% priority sector (agro-based / bio-fertilizer) + 10% new technology; overall cap **₹40 L**, reached at ~₹90 L fixed capital (our ₹155 L fixed capital comfortably maxes it). *Palakkad does not get the extra 10% district bonus (that is only Idukki/Wayanad/Kasargode/Pathanamthitta).*
- **+ AIF** (central): 3% interest subvention up to ₹2 cr loan for 7 yrs + CGTMSE guarantee fee government-paid. Different basis (interest vs capital) → **stacks legitimately with ESS**; confirm convergence in writing with DIC + bank.
- **+ CGTMSE:** collateral-free loan → keeps the farm un-mortgaged.
- **Do NOT use PMEGP** — its rules disqualify units that have availed any other government subsidy, so it would void ESS + AIF. (PMEGP is only the fallback if no bank loan is possible.)
- **Effective term-loan interest:** 9% − 3% = **~6%**.

12. Working capital (indicative, annual)

Feedstock collection/transport, enrichment inputs (poultry manure, neem cake, rock phosphate, microbial consortium), packaging, wages, power, and ~1–2 months finished-goods holding. Phase-1 operating cost ≈ **₹60 L/yr**; Phase-2 ≈ **₹1.05 cr/yr**.

13. Profitability & ROI

Common assumptions: 300 operating days; biochar 1 t ≈ 2.2 t CO_2; carbon net realization ~₹10,000/t CO_2 (after aggregator/MRV); enriched blended price ₹35/kg (base) to ₹40/kg (with-carbon scenario). Effective interest 6%.

Scenario	Biochar	Revenue	Op. cost	Net profit (approx)	Payback
Phase-1 base (physical only)	300 t	₹1.05 cr	₹0.60 cr	**~₹22 L**	~3.5–4 yr
Phase-1 + carbon	300 t	₹1.05 cr + ₹0.66 cr = ₹1.71 cr	₹0.65 cr	**~₹80 L**	~1.6–2 yr
Phase-2 base (physical only)	600 t	₹2.10 cr	₹1.05 cr	**~₹80 L**	—
Phase-2 + carbon	600 t	₹2.4 cr + ₹1.32 cr = ₹3.72 cr	₹1.20 cr	**~₹2.1 cr**	<1 yr

Net capital at risk (Phase-1) ≈ ₹1.30 cr after ESS ₹40 L. **ROI (Phase-1 base): ~17%; with carbon: ~60%.** Carbon revenue is upside — treat physical-only as the bankable base case; carbon requires certification + a signed aggregator offtake.

Drivers: near-free anchor feedstock, ~6% subsidised interest, ₹40 L ESS, and selling enriched (not raw) product.

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14. Carbon-credit strategy & aggregator options

At ~660 t CO_2/yr (Phase-1) / ~1,320 t CO_2/yr (Phase-2), volumes are **too small to transact directly with corporate buyers** (~85% of CORCs are sold via brokers/aggregators). Route through an **aggregator/developer** who provides MRV, certification (Puro.earth CORC methodology) and buyer access.

Party	Role / fit
Varaha (Gurugram/New Delhi) — India's largest biochar CDR developer; Puro.earth + Verra; offtakes with Google, Microsoft, Kellanova	**Best first contact.** Runs the **Varaha Industrial Partners Program (VIPP)** — lets facilities with feedstock + conversion capability use Varaha's digital MRV + credit commercialization and issue CORCs under Puro's biochar methodology.
Equilibrium — industrial biochar operator; 180,000 t offtake with Altitude	Developer/partner for industrial biochar; potential offtake channel.
Eco Saarthi — cooperative model aggregating agri residue → biochar/bio-coal	Fit for a cooperative/aggregation structure.
Puro.earth	The **registry/standard** (CORCCHAR biochar methodology); certification happens through a developer.
BioFlux (advisory)	Independent biochar-CDR advisory — feasibility, feedstock/logistics design, MRV & certification, buyer alignment. Useful consultant to structure the project for certification.
Marketplaces/brokers — **Supercritical, Senken**	Secondary sale/price-discovery channels.

Action: approach **Varaha VIPP first** for an MRV + offtake partnership; engage **BioFlux** (or equivalent) to make the plant EBC/Puro-ready; target a **Letter of Intent (LOI)** before Phase-2 to bank the carbon revenue. Ensure feedstock is documented **residual/waste biomass** (fronds, husk, shells, coir) for additionality.

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15. Value addition — enriched bio-fertilizer

Base biochar is K-rich; enrich to a balanced slow-release organic input: - **Nitrogen/phosphorus:** poultry manure, neem cake, bone meal / rock phosphate. - **Biology (branded premium):** microbial consortium — Azospirillum/Azotobacter (N-fix), PSB/phosphobacteria, potash mobilizers, Trichoderma, mycorrhiza (VAM). - **Base/biology co-compost:** vermicompost + cow dung; Kerala liquid organics (fish amino acid, panchagavya, jeevamrutham); seaweed extract for micronutrients. - **Methods:** nutrient impregnation/soaking then low-temp drying; or co-composting/vermicomposting with ~10% biochar; pelletize to cut leaching. - **Certifications for premium/export:** EBC, IBI, organic-input (Jaivik Bharat / NPOP for export).

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16. Implementation schedule (sequence, not calendar)

1. Feedstock lab proximate analysis (moisture, fixed carbon, ash, K) — decides final sizing.
2. Udyam registration; DIC/ESS pre-consultation; feedstock MoUs.
3. Land-use confirmation; **CTE (KSPCB, rated 2 t/day)**.
4. Machinery quotes & order (reactor 1 t/day; gas train/dryer/civil for 2 t/day).
5. Civil construction + plant erection with reactor #2 tie-ins pre-provisioned.
6. Panchayat D&O + building permit + fire NOC; **CTO**.
7. AIF term loan + CGTMSE; commissioning; ESS claim on installed assets.
8. Product validation on own farm; EBC/organic certification.
9. Engage carbon aggregator (Varaha VIPP); secure LOI.
10. Phase-2: add reactor + scale feedstock aggregation.

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17. Risks & mitigation

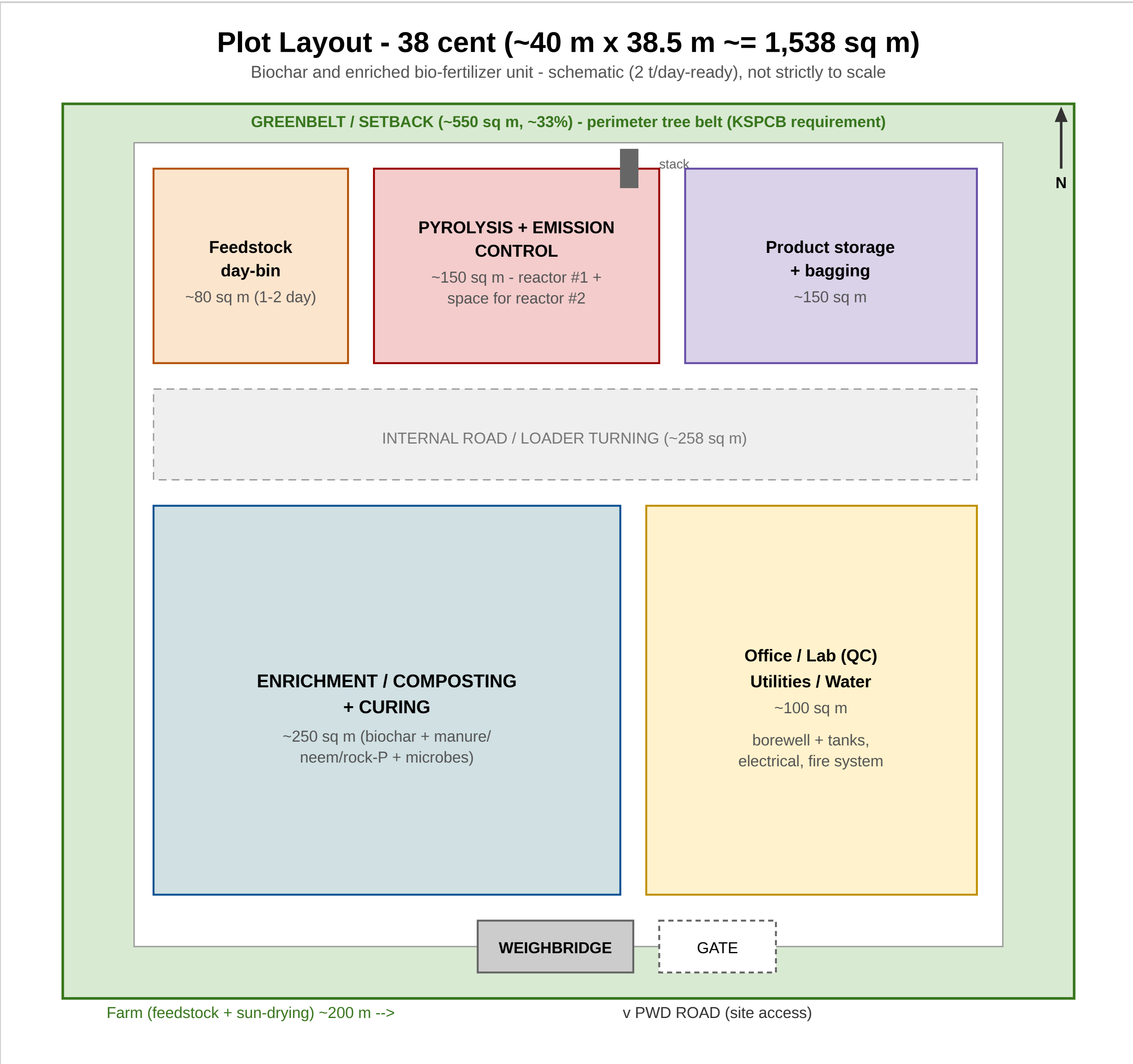
Risk	Mitigation
Feedstock moisture / drying energy	Sun-dry at farm; waste-heat dryer; lab-test first
Demand-side weakness (Indian biochar)	Own-farm consumption; sell enriched/branded; export the credit
Carbon-credit dependency	Bank case = physical only; carbon = upside pending LOI
Regulatory sequencing	CTE (2 t/day) → build → CTO; consultant support
Feedstock security	Supply MoUs (partner + local sources)
Product consistency for certification	Continuous reactor + fixed blend recipe

18. Assumptions & disclaimer

All costs, yields, prices and returns are **planning-grade indicative estimates** (mid-2026 market references) for the purpose of scheme application and bank appraisal. They must be confirmed by: (a) a laboratory proximate analysis of the actual feedstock; (b) formal machinery quotations; (c) written confirmation of ESS/AIF eligibility and convergence from DIC Palakkad and the lending bank; and (d) a carbon-credit LOI from an aggregator before carbon revenue is counted. Regulatory categories and subsidy rates are subject to the prevailing KSPCB, DIC and scheme guidelines at the time of application.

Annexure A — Plot layout (38 cent, 2 t/day-ready)

A scaled schematic is provided as `plot-layout.svg` (open in any browser to print on one page). Text schematic below for quick reference:

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                                       N ^
============ GREENBELT / SETBACK  ~550 m² (perimeter tree belt) ============
|                                                             [stack]        |
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|  +-------------+   +----------------------+   +-------------------------+|
|  | Feedstock   |   | PYROLYSIS + EMISSION |   | Product storage         ||
|  | day-bin     |   | CONTROL   ~150 m²    |   | + bagging   ~150 m²     ||
|  | ~80 m²      |   | reactor #1 (+ space  |   |                         ||
|  | (1-2 day)   |   |  for reactor #2)     |   |                         ||
|  +-------------+   +----------------------+   +-------------------------+|
|  - - - - - - INTERNAL ROAD / LOADER TURNING (~258 m²) - - - - - - - - - - |
|  +---------------------------+   +-------------------------------------+ |
|  | ENRICHMENT / COMPOSTING   |   | Office / Lab (QC)                   | |
|  | + CURING     ~250 m²      |   | Utilities / Water   ~100 m²         | |
|  | (char + manure/neem/      |   | borewell+tanks, electrical, fire    | |
|  |  rock-P + microbes)       |   |                                     | |
|  +---------------------------+   +-------------------------------------+ |
|                         [ WEIGHBRIDGE ]   [ GATE ]                        |
==========================================================================
          ▼ PWD ROAD (access)        → Farm (feedstock + sun-drying) ~200 m
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Notes: greenbelt on all sides for KSPCB consent; pyrolysis block sized with vacant bay + pre-laid foundation/tie-ins for **reactor #2** (Phase-2); large biomass storage is kept **off-site at the farm** (200 m), so only a 1–2 day day-bin sits on the plot; weighbridge and gate at the road frontage.

Annexure B — Term-loan amortisation schedule

Assumptions: Term loan **₹110 L**; effective rate **6% p.a.** (9% − 3% AIF interest subvention); tenure **7 years**; **6-month construction moratorium** (interest served, no principal in Year 1); **Kerala ESS ~₹40 L received ~Month 12 and applied as principal prepayment**, leaving ~₹70 L to amortise over Years 2–7. Equal annual instalment (EAI) Years 2–7 ≈ **₹14.24 L** (≈ **₹1.19 L/month EMI**).

Year	Opening (₹L)	Interest @6% (₹L)	Principal repaid (₹L)	ESS prepayment (₹L)	Closing (₹L)
1	110.00	6.60	0.00 (moratorium)	40.00	70.00
2	70.00	4.20	10.04	–	59.96
3	59.96	3.60	10.64	–	49.32
4	49.32	2.96	11.28	–	38.04
5	38.04	2.28	11.96	–	26.08
6	26.08	1.56	12.68	–	13.40
7	13.40	0.80	13.44	–	~0.00
Total		**~22.0**	**~70.0**	**40.0**	

AIF benefit: the 3% subvention saves ~**₹11 L** of interest over the loan life versus a 9% loan on the same balances. Guarantee fee is government-paid under AIF (CGTMSE), so the loan is **collateral-free** — the farm is not mortgaged.

Annexure C — Month-by-month cash flow, Year 1 (Phase-1 ramp)

Assumptions (Phase-1, physical-sales base case): commissioning at **Month 7**; ramp 25% → 100% by Month 12. Full-capacity monthly revenue **₹8.75 L** (₹1.05 cr/yr); operating cost ≈ **₹2.0 L fixed + ₹3.0 L variable at 100%**; interest served during moratorium ≈ **₹0.55 L/month**. Capex is funded by equity + term loan (not shown in operating flow). ESS ₹40 L inflow at Month 12 is a financing item applied to loan prepayment (Annexure B).

Month	Phase	Capacity	Revenue (₹L)	Op. cost (₹L)	EBITDA (₹L)	Interest (₹L)	Net op. cash (₹L)
1–6	Construction / pre-op	–	0.00	trial only	–	0.55/mo	interest served from loan
7	Commissioning	25%	2.19	2.75	−0.56	0.55	−1.11
8	Ramp	40%	3.50	3.20	0.30	0.55	−0.25
9	Ramp	55%	4.81	3.65	1.16	0.55	0.61
10	Ramp	70%	6.13	4.10	2.03	0.55	1.48
11	Ramp	85%	7.44	4.55	2.89	0.55	2.34
12	Full	100%	8.75	5.00	3.75	0.55	3.20
H2 total			**32.82**	**23.25**	**9.57**	**3.30**	**6.27**

Steady state (Year 2+, Phase-1 full): monthly revenue ₹8.75 L − opex ₹5.0 L = **EBITDA ₹3.75 L**; less EMI ₹1.19 L → **net cash ~₹2.56 L/month (~₹30.7 L/yr)** before tax and depreciation.

Debt-service coverage (Year 2): EBITDA ≈ ₹45 L ÷ debt service ≈ ₹14.24 L → **DSCR ≈ 3.1** (comfortably bankable; lenders typically want ≥ 1.5–2.0).

Upside: adding realized carbon revenue (~₹0.66 cr/yr in Phase-1, ~₹1.32 cr/yr in Phase-2) lifts EBITDA sharply and shortens payback to under ~2 years — kept out of the base case above pending a signed aggregator LOI.

Annexure D — Machinery quotation-comparison template (fillable)

How to use: send this to at least **two local vendors (Coimbatore/Pollachi hub)** and **one national continuous-technology vendor**. Ask each to quote the **Phase-1 reactor (1 t/day)** together with the **2 t/day-sized gas train and dryer** (so no re-work at expansion), and to price reactor #2 tie-in provisions as a separate option. Enter the received figures in Table D-2, then carry the sub-totals into **Section 10 — Cost of project** (mapping shown in the "→ Cost item" column).

Table D-1 — Scope & required specification (issue to vendors)

#	Equipment	Sized for	Required specification / notes	→ Cost item
1	Pyrolysis / carbonization reactor	**Phase-1: 1 t/day** (~200–250 kg/hr feed)	Continuous or semi-continuous; multi-feedstock (briquette, husk, frond, shell, coir); yield ≥30% dry basis	#4
2	Reactor #2 tie-in provisions	Phase-2 (option)	Pre-laid foundation bolts, flanged feed/gas stubs, spare panel ways — quote as separate option	#4 (future)
3	Feed system (hopper, screw/belt conveyor, metering)	**2 t/day (common header)**	Variable-speed feed; suited to briquettes + shredded biomass	#4
4	Waste-heat dryer	**2 t/day**	Uses pyrolysis/syngas waste heat; inlet moisture up to ~25%	#5
5	Emission control train — afterburner/flare + cyclone + alkaline scrubber + mist eliminator + ID fan + stack	**2 t/day**	Must meet **KSPCB (Orange) ambient/stack norms**; syngas combustion; vendor to state guaranteed emission values	#5
6	Char cooling / quench + discharge	1 t/day	Sealed cooling; moisture/temperature controlled	#5
7	Sieving / screening	1 t/day	Grade separation (powder/granular)	#6
8	Enrichment mixer / blender	1 t/day	For char + manure/neem/rock-P + microbial slurry	#6
9	Pelletizer (optional)	1 t/day	For densified enriched product	#6
10	Weighing + bagging	1 t/day	25 kg bags + FIBC jumbo option	#6
11	Control panel / instrumentation (PLC/SCADA)	2 t/day	Temp/pressure monitoring, interlocks, e-stop	#7
12	Erection, commissioning & operator training	—	Include site supervision & trial run	#4–#7
13	Freight, insurance, unloading	—	To site, Eruthenpathy	#4–#7
14	Taxes (GST)	—	State applicable rate separately	—

Table D-2 — Price comparison (fill from vendor quotes, ₹)

#	Equipment (as above)	Sized for	Vendor A: ____	Vendor B: ____	Vendor C: ____
1	Pyrolysis reactor	1 t/day	₹___	₹___	₹___
2	Reactor #2 tie-in provisions (option)	Phase-2	₹___	₹___	₹___
3	Feed system	2 t/day	₹___	₹___	₹___
4	Waste-heat dryer	2 t/day	₹___	₹___	₹___
5	Emission control train	2 t/day	₹___	₹___	₹___
6	Char cooling / quench	1 t/day	₹___	₹___	₹___
7	Sieving / screening	1 t/day	₹___	₹___	₹___
8	Enrichment mixer	1 t/day	₹___	₹___	₹___
9	Pelletizer (optional)	1 t/day	₹___	₹___	₹___
10	Weighing + bagging	1 t/day	₹___	₹___	₹___
11	Control panel / SCADA	2 t/day	₹___	₹___	₹___
12	Erection, commissioning, training	—	₹___	₹___	₹___
13	Freight, insurance	—	₹___	₹___	₹___
	Sub-total (ex-GST)		₹___	₹___	₹___

#	Equipment (as above)	Sized for	Vendor A: ____	Vendor B: ____	Vendor C: ____
14	GST	—	₹____	₹____	₹____
	TOTAL (incl. GST)		**₹____**	**₹____**	**₹____**
	Carry to Section 10: items #4 (reactor+feed) + #5 (dryer+emission+cooling) + #6 (sieving/mixing/bagging) + #7 (electrical/control)				

Table D-3 — Commercial-terms comparison (fill from vendor quotes)

Parameter	Vendor A	Vendor B	Vendor C
Make / model offered			
Rated capacity (kg/hr feed; t/day biochar)			
Guaranteed biochar yield (% dry basis)			
Material of construction (reactor)			
Connected electrical load (kW)			
Guaranteed stack emissions vs KSPCB norms (state values)			
Feedstock types accepted (briquette/husk/frond/shell/coir)			
Delivery period (weeks from PO)			
Warranty (months)			
After-sales / spares — distance from site			
Payment terms			
Training & commissioning included?			
References / plants in operation (contactable)			

Evaluation guidance: do not select on price alone — weight **emission-compliance (for CTO), guaranteed yield, feedstock flexibility, and after-sales proximity** (favouring the Coimbatore/Pollachi hub for fast service). Confirm the quoted gas train and dryer are genuinely **rated for 2 t/day** so Phase-2 needs only the second reactor module.